

Beyond Isolation: The Evidentiary Burden of Generalization

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Abstract

Scientific progress has largely been built on the ability to isolate phenomena — controlling variables, enabling repeatable observation, reducing complexity to a tractable question. That success has produced an implicit assumption: that a result which behaves consistently under isolated conditions is generally assumed to possess broader validity unless later contradicted. This paper treats that assumption as worth making explicit rather than leaving it to operate unexamined. It distinguishes isolation from generalization as two different scientific achievements rather than sequential steps of one process, and proposes that the legitimacy of extending an isolated finding beyond its original conditions should itself be treated as an ongoing empirical question rather than a one-time logical inheritance. The paper locates this claim honestly against four existing traditions that already make versions of it — Campbell and Stanley’s internal/external validity distinction, Cartwright’s nomological machines, Steel’s comparative process tracing, and Yarkoni’s generalizability crisis — and argues that what remains genuinely additive is not the diagnosis but a candidate graduated marker for telling well-earned generalization from poorly-earned generalization: *degree of coherence*, built by combining Wimsatt’s robustness-through-independent-derivation with Kahneman and Klein’s adversarial collaboration method, and explicitly bounded by Orzack and Sober’s critique that convergence across methods can reflect a shared blind spot rather than a real pattern. The marker is tested against two real, documented cases in both directions — the 2023 adversarial collaboration that reconciled Kahneman and Deaton’s income-happiness plateau with Killingsworth’s contradicting experience-sampling data, and a 2015 preregistered adversarial collaboration that failed to replicate an apparently well-converged eye-movement-memory effect, consistent with the shared-blind-spot failure mode Orzack and Sober describe. The paper is offered as a navigational extension of the standard established in Paper 2 — not a replacement for existing philosophy of science, but a proposed third activity, alongside discovery and generalization, for how understanding should continue to evolve as reality itself continues to evolve.

Keywords: generalization, extrapolation, evidentiary burden, robustness, adversarial collaboration, external validity, navigational coherence

I. The Central Question

Scientific progress has largely been built on the ability to isolate phenomena. Isolation reduces complexity. It allows variables to be controlled. It enables repeatable observation. Its success has been extraordinary.

Because of this success, an implicit assumption has gradually emerged: if a phenomenon behaves consistently under isolated conditions, the resulting explanation is generally assumed to possess broader validity unless contradicted by later evidence.

This paper asks whether that assumption should remain implicit.

The central question is not whether isolation has been useful — it unquestionably has — but whether the legitimacy of extending isolated findings beyond their original conditions should itself become an explicit scientific question, carrying its own evidentiary burden rather than inheriting validity automatically from the isolated result that produced it.

Isolation and generalization are, on this account, different scientific acts. Isolation answers one question: what happens under these specific conditions? Generalization answers a different one: under what wider conditions should this result continue to hold? Historically these two questions have often been treated as successive steps of one process. This paper treats them instead as distinct scientific achievements, each carrying its own standard of evidence.

II. Isolation as Local Validity

Isolation may be understood as establishing local validity. A result obtained under isolation is rigorously valid within the conditions under which it was demonstrated. Whether that validity extends beyond those conditions remains an independent scientific question. This does not weaken isolated science. It clarifies its scope.

The Newtonian mechanics case. For nearly all everyday engineering, Newtonian mechanics is not merely adequate — it is exactly right, to any precision an engineer would ever need: bridge construction, ballistics, orbital calculations for most spacecraft, structural load-bearing. The theory succeeds because its domain of application — moderate speeds, moderate gravitational fields — is the domain almost all engineering actually occurs in.

Yet no physicist would conclude from this success that Newtonian mechanics describes reality universally. At speeds approaching light, or in the presence of extreme gravitational fields, its predictions diverge measurably from what is observed, and general relativity is required instead.

Crucially, nobody says Newton was wrong. His mechanics is not a discredited theory awaiting replacement — it is a rigorously valid local theory, whose domain of validity is now precisely known, because someone did the harder work of finding where it stops holding rather than assuming it holds everywhere by default. Local usefulness is not, by itself, evidence of universal validity — and finding the actual boundary of that validity is not a mark against a theory. It is scientific maturity that a purely successful, unchallenged theory never gets forced to develop.

III. A Proposed Expansion of Evidentiary Burden

This paper proposes that the burden associated with scientific generalization should be made explicit. Rather than assuming that validity naturally accompanies extrapolation, broader claims should continuously demonstrate coherence as additional relational conditions are introduced. The question becomes: does the explanatory framework continue to remain coherent as progressively larger portions of reality are included? If coherence remains, its domain of validity expands. If coherence deteriorates, its domain of validity contracts. Generalization therefore becomes an ongoing empirical achievement rather than a one-time logical inheritance.

This proposal aligns naturally with domains already organized around continuous recalibration — weather forecasting, navigation, ecological management, medical diagnosis, economic systems. None of these lose scientific legitimacy because they require continual recalibration; their continual recalibration is part of their scientific rigor. Under this proposal, broader scientific generalizations would carry a similar evidentiary responsibility.

Scientific maturity might therefore be understood along two complementary dimensions. **Internal validity** asks whether the isolated phenomenon was rigorously established under its stated conditions — the great achievement of classical scientific methodology. **External coherence** asks whether the explanatory framework continues to maintain coherence across larger scales, interacting systems, temporal development, environmental variation, and newly incorporated evidence, without accumulating unresolved contradiction. Neither dimension replaces the other; both contribute to scientific confidence.

IV. Prior Art — Locating This Paper Honestly

The core distinction above is not new, and treating it as new would be the first thing a knowledgeable reviewer would correct. It already exists, independently, in at least four traditions.

Campbell and Stanley (1963), in the foundational text on experimental design, name internal and external validity as exactly this split — whether a causal claim holds within a study, versus whether it generalizes beyond it. This is standard methodology-course material, decades old.

Nancy Cartwright, in *The Dappled World* (1999), makes a version of this argument at the level of metaphysics rather than method: scientific laws hold only within “nomological machines” — specific, bounded arrangements of stable conditions — and exporting a law beyond the machine that produced it requires independent justification, not automatic inheritance. Cartwright’s image of a dappled, patchwork world of local, true laws rather than one universal fabric is close to this paper’s own claim about explanatory frameworks and their domains.

Daniel Steel, in *Across the Boundaries: Extrapolation in Biology and Social Science* (2007), addresses this problem directly and technically: when is it legitimate to extend a causal finding — from an animal model to humans, from a pilot programme to a population — and proposes comparative process tracing as a method, checking whether the mechanism producing an effect actually exists in the new target domain before assuming the finding travels.

Tal Yarkoni, in “The Generalizability Crisis” (*Behavioral and Brain Sciences*, 2022), makes the most directly relevant and most recent case: an extensively discussed, still-contested target article arguing psychology routinely makes sweeping generalizations from narrow experiments without the statistical grounds to support them. Critics proposed competing frameworks in response — this is a live, unresolved debate in a real field, not settled ground.

What that means for this paper: the diagnosis — isolation and generalization are different acts, and the second one has been getting a free pass — is not this paper’s original contribution. What is genuinely additive is connecting that well-documented problem to a working operational marker for handling it, rather than only naming it again. The traditions above are strong on diagnosis and comparatively thin on infrastructure — Steel’s comparative process tracing is the closest to a practical method, and it is narrower than what this research programme’s own tools already do in practice. If this paper has a real claim to make, it is there.

V. Degree of Coherence — A Graduated, Testable Marker

The argument so far has a real gap: as stated, it risks implying that all generalized claims carry the same ongoing evidentiary burden, indefinitely. That would prove too much. Conservation of energy has survived an enormous, continually expanding range of genuinely severe tests across wildly different domains for over a century. A single social-science finding, casually stretched past the one population it was tested on, has survived almost none. Treating both as equally suspect flattens a real and important difference, and risks handing a legitimate-sounding argument to people who would misuse “nothing generalizes for free” as cover for rejecting well-earned science along with poorly-earned science. What is needed is a marker that can tell the two apart, not just a demand that both keep proving themselves forever in the same way.

Two existing, independently developed methods, put together, supply a candidate marker.

Robustness through independent derivation. William Wimsatt’s “Robustness, Reliability, and Overdetermination” (1981) argues that confidence in a result should scale with the number of genuinely independent means used to derive it — different methods, different disciplinary assumptions, different fields converging on the same conclusion despite having no reason to agree by default. Richard Levins’s 1966 formulation is the memorable version: “the truth is at the intersection of independent lies” — each method is imperfect in its own particular way, and where independently imperfect methods nonetheless agree, that agreement is worth taking seriously as evidence something real is being pointed at. This gives cross-field validity a precise operational meaning: not more confirming instances from the same method, but convergence across methods that have no shared reason to fail in the same way.

Adversarial collaboration. Kahneman and Klein’s “Conditions for Intuitive Expertise: A Failure to Disagree” (*American Psychologist*, 2009) supplies a formal method for the second half of the marker: two researchers who genuinely disagreed about a question, working jointly for years, agreeing in advance on what conditions would need to hold for either position to be vindicated, and keeping the disagreement itself visible in the published result rather than smoothing it into a tidier joint conclusion. This is the real-world version of the strongest avail-

able dissent being built into the test itself rather than merely addressable afterward — and it is already a live instance of this research programme’s own criteria-for-success requirement, since the method’s whole discipline is agreeing on the test before running it.

Proposed marker. A claim’s degree of coherence increases specifically when it survives both tests together — independent, cross-field triangulation (Wimsatt), and adversarial collaboration against its strongest available opposing case, with that opposition preserved rather than resolved away (Kahneman and Klein) — rather than from either alone, and rather than from simple accumulation of same-method confirming instances. This gives generalization a graduated scale rather than a flat yes/no, and gives conservation of energy and a single social-science finding principled, different positions on it, rather than treating them alike.

The honest limit, stated plainly. Wimsatt’s own tradition has a serious internal critique. Orzack and Sober (1993) argued that robustness analysis can be a non-empirical illusion — several methods can converge not because they are independently tracking something real, but because they silently share a common, unexamined assumption, in which case their agreement confirms nothing about the world and everything about the shared blind spot. Any use of degree of coherence as a marker has to take this seriously: convergence across methods is suggestive, not conclusive, and the burden of checking whether the convergence is genuinely independent — rather than several instances of the same hidden assumption — falls on whoever is claiming the coherence, not on whoever is skeptical of it.

VI. Worked Cases — The Marker Tested in Both Directions

The degree-of-coherence marker proposed in Section V is only useful if it can discriminate — if it treats a claim that has actually earned convergence differently from a claim that only appears to have. Two real, documented cases test it in both directions: one where independent triangulation and adversarial collaboration together produced a more coherent, more precise claim; one where apparent convergence across studies turned out, under the same kind of scrutiny, to be the shared blind spot Orzack and Sober warned about rather than evidence of anything real.

A positive case: income and emotional well-being. Kahneman and Deaton (2010), analysing retrospective day-recall survey data from roughly 450,000 respondents, reported that emotional well-being rises with log-income up to a threshold near \$75,000 per year, after which it plateaus. Killingsworth (2021), using a genuinely independent method — real-time experience sampling via smartphone, rather than retrospective recall — found no such plateau: happiness continued rising with log-income with no flattening in sight. These were not minor discrepancies at the edges of two compatible studies; they were a flat contradiction between two careful, well-powered analyses using different instruments on different samples.

Rather than trading rebuttals, Kahneman and Killingsworth carried out a formal adversarial collaboration, joined by Barbara Mellers as a third party: pooling raw data, agreeing on the reanalysis in advance, and publishing regardless of outcome, in the spirit of the method Kahneman and Klein had themselves formalised in 2009. The resolution was not that one side had simply been right. A reanalysis of Killingsworth’s experience-sampling data confirmed

the flattening pattern, but only for the least happy respondents; for everyone happier than that, well-being continued rising steadily with income, and even accelerated for the happiest group. The apparent contradiction dissolved once the two studies' results were read as describing different parts of the same more complex, heterogeneous relationship — a relationship neither original study, working alone, had the resolution to detect (Killingsworth, Kahneman, & Mellers, 2023).

This is the marker firing as proposed. Independent triangulation surfaced a real disagreement rather than papering over it; adversarial collaboration, with the strongest available opposing case built into the test itself, produced a claim more coherent and more precise than either original finding — not because either side conceded, but because the joint reanalysis found what neither had been positioned to see. Consistent with this paper's own framing in Section III, the achievement did not close the question permanently: published exchanges since 2023 have continued to probe the reconciled finding's causal assumptions, which is itself the ongoing-recalibration signature this paper argues generalization should carry, rather than a flaw in the case.

A cautionary case: eye movements and memory retrieval. A separate literature had, by the early 2010s, accumulated what looked like convergent support across several independent studies for Saccade-Induced Retrieval Enhancement — the claim that brief horizontal eye movements improve free recall of episodic memories, with implications for EMDR-style therapeutic technique. On Wimsatt's own logic, several studies from different groups agreeing on an effect is exactly the pattern that should raise confidence.

Matzke, Nieuwenhuis, van Rijn, Slagter, van der Molen, and Wagenmakers (2015) tested this directly, running a preregistered adversarial collaboration between proponents and skeptics of the effect: the two sides agreed on the experimental design, the analysis plan, and the criteria for interpreting the result before any data were collected, with an independent referee to resolve procedural disputes. The eye-movement effect did not replicate. The authors' own explanation is the one this paper's Section V already names as the honest limit on robustness reasoning: the earlier convergence across studies may have reflected shared questionable research practices — common to the whole cluster of prior work — rather than several independent methods genuinely tracking a real effect. What had looked like cross-study robustness was, on the more demanding joint test, indistinguishable from Orzack and Sober's shared blind spot.

What the pair together shows. Neither case alone would be enough. The income case alone would read as a survivorship-biased success story, chosen because it worked. The eye-movement case alone would read as a case against the marker rather than a test of it. Together, they show the marker doing the discriminating work Section V asks of it: rewarding a claim that survived contact with independently-derived contradiction and organised adversarial scrutiny, and correctly failing to reward a claim whose apparent convergence did not survive the same test. This is not, on two cases, a systematic validation of the marker — that would require a deliberately assembled sample of candidate cases selected in advance, not two cases located because they illustrated the point well. It is evidence that the marker can discriminate at least once in each direction, which is the minimum a proposal of this kind owes before asking to be taken seriously as more than a redescription of intuition.

VII. Implications for Navigational Theory

If developed further, this distinction represents an extension of Navigational Theory rather than a replacement of existing philosophy of science. The proposal suggests three distinct scientific activities: **discovery** — what happens under specified conditions; **generalization** — under what additional conditions does this continue to hold; and **navigation** — how should understanding continue to evolve as reality itself continues to evolve. Rather than treating validity as inherited automatically through extrapolation, validity becomes something accumulated through continued coherence across expanding relational domains, with degree of coherence offering a candidate way to make that accumulation gradable rather than merely asserted.

VIII. Scope and Limitations

This paper's contribution is narrower than it might first appear, and stating the boundary plainly matters more than claiming open territory.

The diagnosis is not new; the marker is the only candidate contribution. Section IV names this directly. Four independent traditions already distinguish isolation from generalization. What this paper adds is a specific way of grading the space between them — and that addition rests entirely on combining two existing methods, not inventing either.

The two worked cases in Section VI are illustrative, not a systematic test. They were located because they cleanly demonstrate the marker discriminating in both directions, not selected from a pre-specified sample of candidate cases screened in advance. A systematic review — assembling every documented instance where independent triangulation and formal adversarial collaboration both occurred on the same claim, then checking the marker's discrimination against all of them — would be needed before “the marker works” could be asserted with confidence rather than illustrated. That review is future work, not completed here.

Orzack and Sober's critique remains live even in the cases used here. The income case's own reconciled finding has already drawn published methodological criticism; the marker having fired positively once is not a guarantee the underlying convergence will hold up indefinitely, which is consistent with this paper's own claim that generalization is an ongoing achievement rather than a settled state.

The proposal remains exploratory. Its value will ultimately depend on whether it produces better scientific navigation across open, adaptive, and interconnected phenomena than existing methodological assumptions alone — a test this paper sets up and partially illustrates, but does not claim to have completed.

IX. Conclusion

Isolation and generalization are different scientific acts, licensed by different evidence. The first has an extraordinarily successful methodology behind it. The second has largely been

assumed to follow from the first automatically, an assumption that several independent traditions have already identified as unearned. This paper's contribution is not that diagnosis — it is a candidate answer to what should replace unearned inheritance: a graduated marker, built from two existing and independently developed methods, that can distinguish generalizations that have done the work of proving they travel from generalizations that have merely been assumed to. Tested against two real cases — one where the marker rewarded a claim that survived independent contradiction and organised adversarial scrutiny, one where it correctly withheld that reward from a claim whose apparent convergence did not survive the same test — the marker discriminates at least once in each direction. The paper does not claim to have proven the marker works in general. It claims to have named, honestly, both what the marker would need to show and what it has so far been shown to do.

Declarations

Author Affiliation and Research Programme

Independent researcher. Formal affiliation: ARTMA ApS. This paper is part of the Audited State Research Programme (auditedstate.app), which investigates Navigational Coherence — how understanding stays sufficiently coherent to support orientation, judgment, revision, and action as evidence, interpretations, and circumstances change — through a network of independently assessable papers rather than a linear series. This paper proposes an extension of the methodological standard established in Navigational Theory, and does not depend on any other paper in the programme for its argument; no cross-citation to other programme papers has been used, consistent with the programme's standing convention for standalone methodological work.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

This paper was developed with the assistance of AI systems as a research and editorial tool, for drafting support, literature cross-referencing, and structural revision. The theoretical framework, arguments, and conclusions are the author's own.

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